In [33]:
```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_style("whitegrid")
```

In [34]:
```
import pandas as pd
df =pd.read_csv("titanic.csv")
df.head()
```

Out[34]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500

In [35]:
```
df.shape
```

Out[35]:
```
(891, 12)
```

In [36]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 118.9 KB
```

In [37]:
```
df.describe()
```

Out[37]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Far
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In [38]:
```
df.describe(include="object")
```

```
C:\Users\DELL\AppData\Local\Temp\ipykernel_22756\702825166.py:1: Pandas4Warning: For b
ackward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype
is specified. This behavior is deprecated and will be removed in a future version. Exp
licitly pass 'str' to `include` to select them, or to `exclude` to remove them and sil
ence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migratio
n-select-dtypes for details on how to write code that works with pandas 2 and 3.
  df.describe(include="object")
```

Out[38]:

	Name	Sex	Ticket	Cabin	Embarked
count	891	891	891	204	889
unique	891	2	681	147	3
top	Braund, Mr. Owen Harris	male	347082	G6	S
freq	1	577	7	4	644

In [39]:
```
df["Survived"].value_counts()
```

Out[39]:
```
Survived
0    549
1    342
Name: count, dtype: int64
```

In [40]:
```
df["Sex"].value_counts()
```

Out[40]:
```
Sex
male      577
female    314
Name: count, dtype: int64
```

In [41]:
```
df["Pclass"].value_counts()
```

Out[41]:
```
Pclass
3    491
1    216
2    184
Name: count, dtype: int64
```

In [42]:
```
df["Embarked"].value_counts()
```

Out[42]:
```
Embarked
S    644
C    168
Q     77
Name: count, dtype: int64
```

In [43]:
```
df.isnull().sum()
```

Out[43]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [44]:
```
df["Age"] = df["Age"].fillna(df["Age"].median())
```

In [45]:
```
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
```

In [46]:
```
df["Cabin"] = df["Cabin"].fillna("Unknown")
```

In [47]:
```
df.duplicated().sum()
```

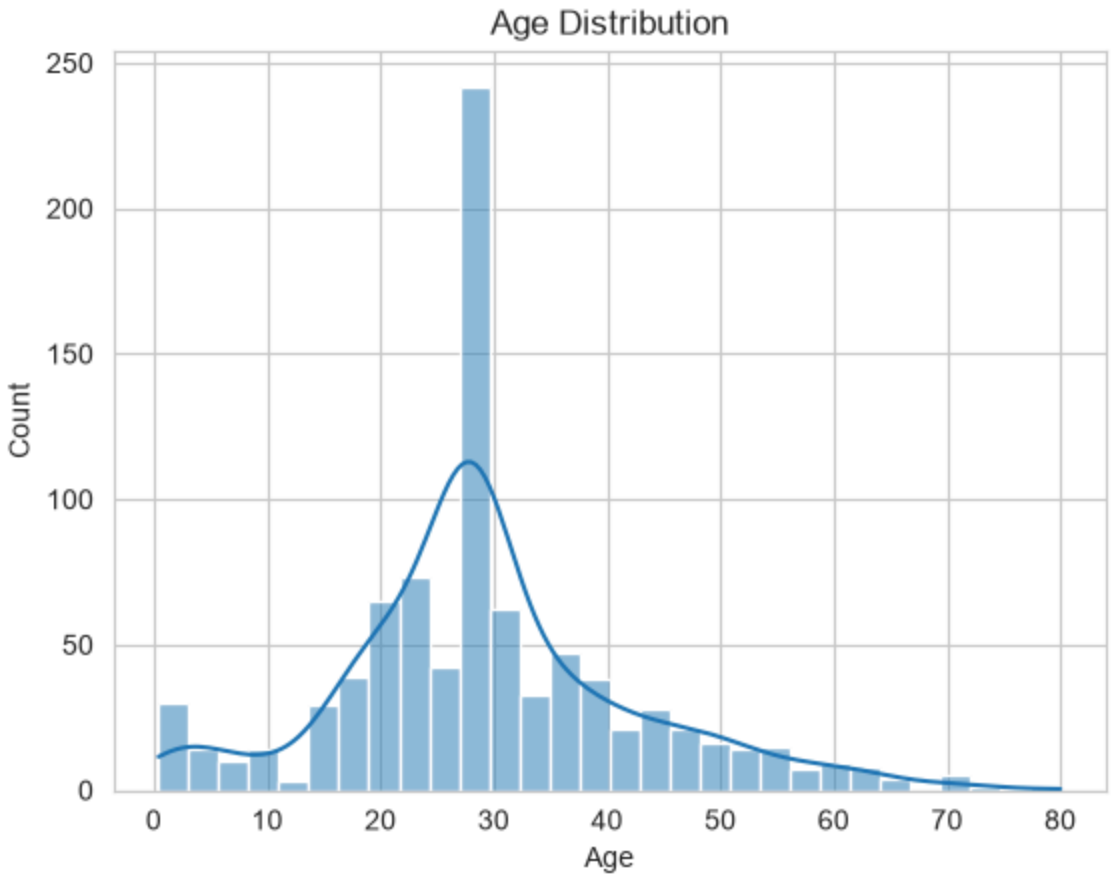
Out[47]:
```
np.int64(0)
```

In [48]:
```
df.isnull().sum()
```

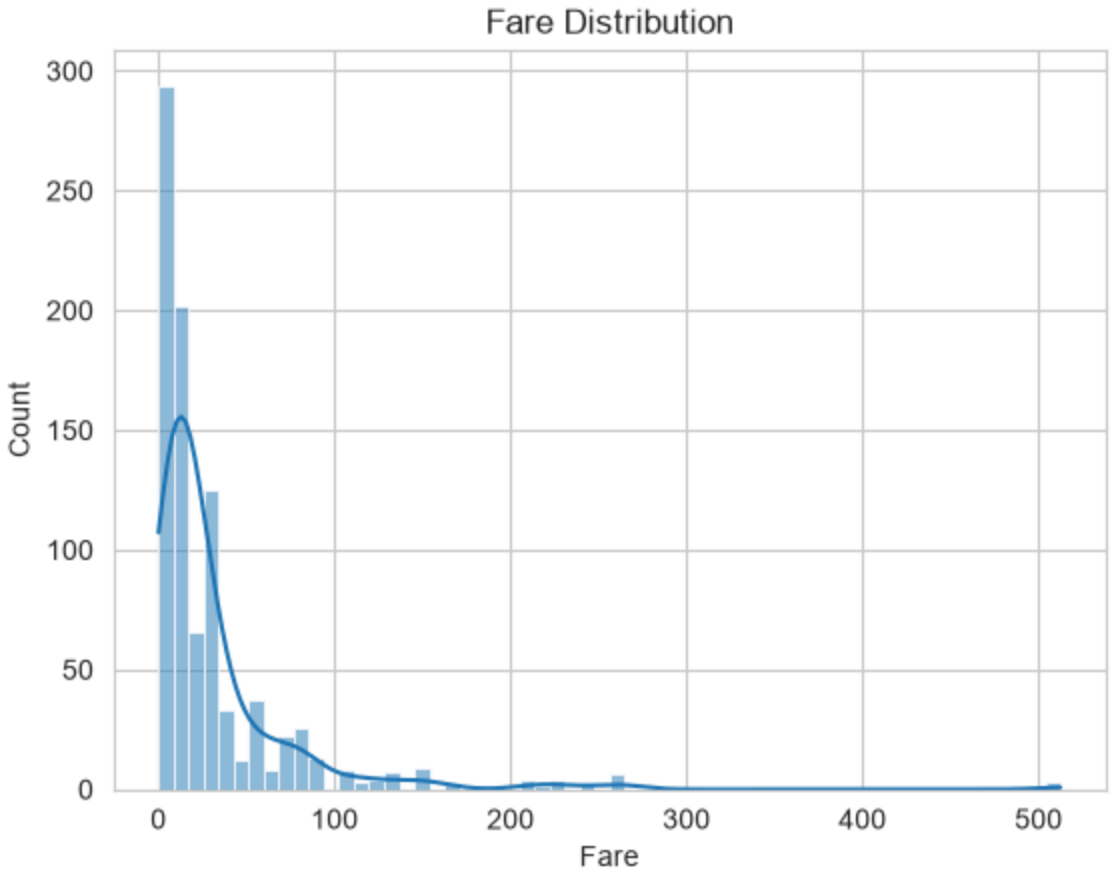
Out[48]:
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          0
Embarked       0
dtype: int64
```

In [49]:
```
df["Has_Cabin"] = (df["Cabin"] != "Unknown").astype(int)
```

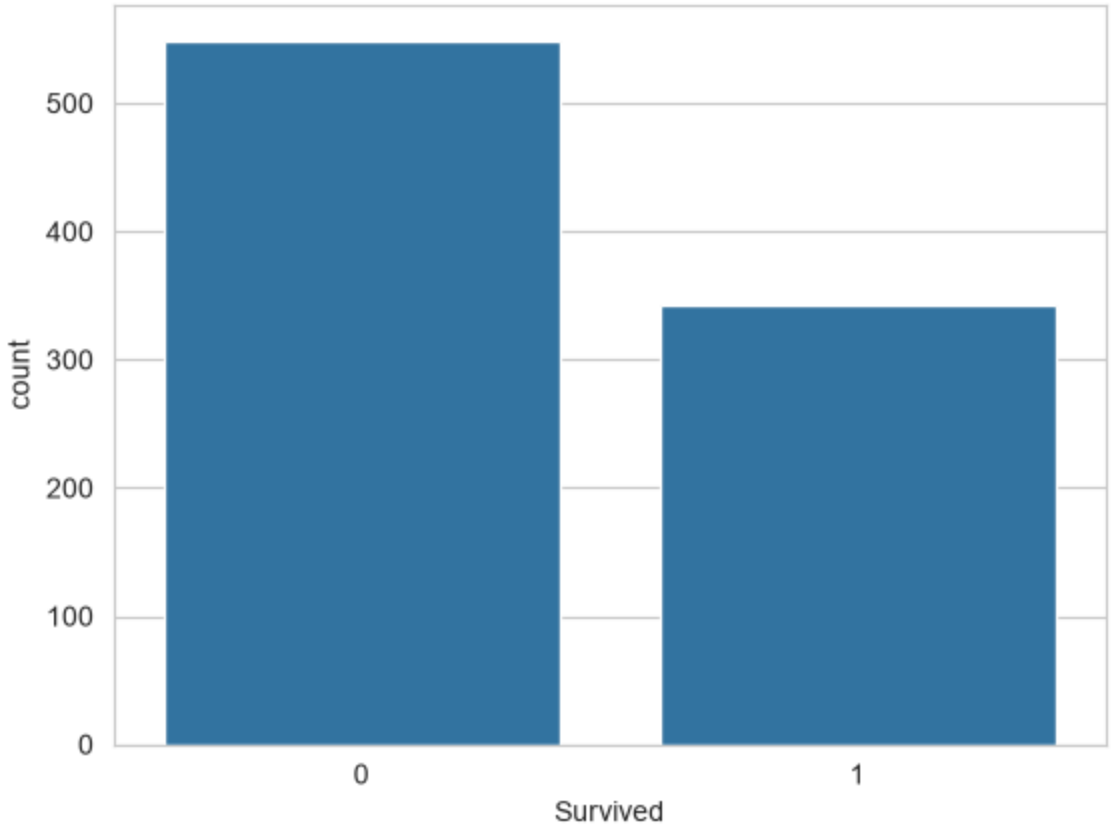
In [50]:
```
sns.histplot(df["Age"], kde=True); plt.title("Age Distribution"); plt.savefig("images
```



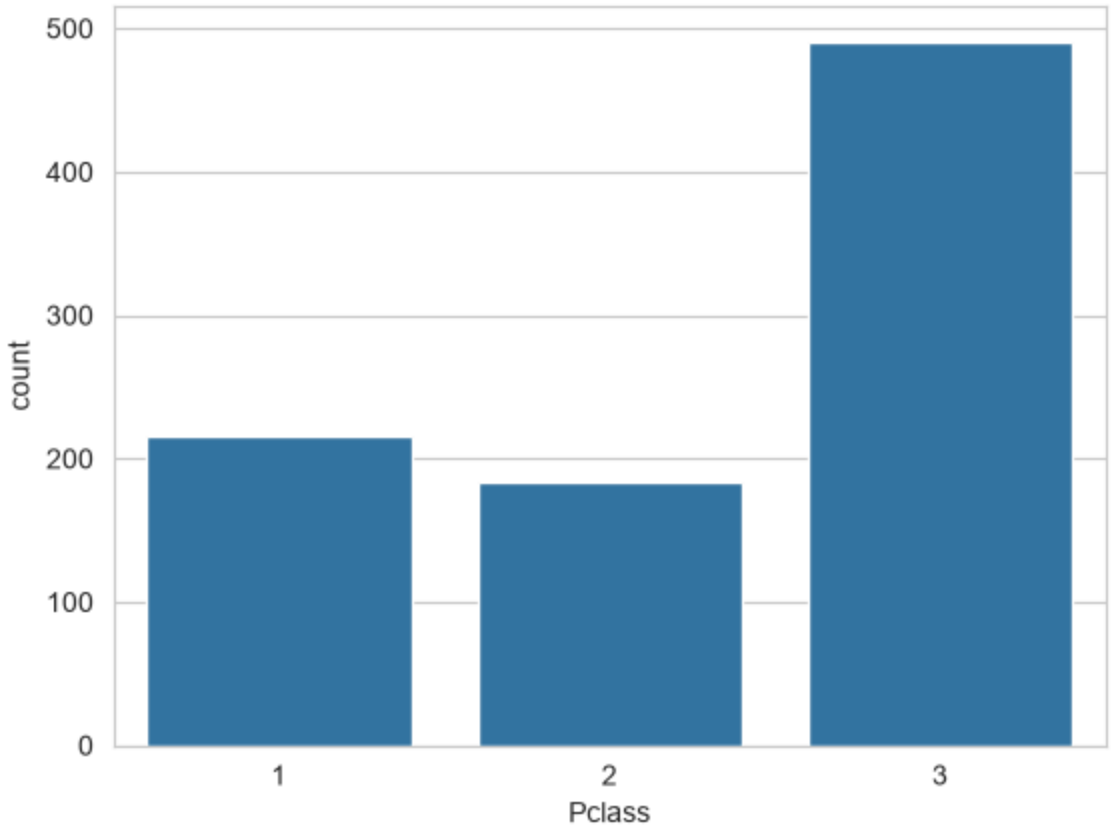
In [51]:
```
sns.histplot(df["Fare"], kde=True); plt.title("Fare Distribution"); plt.savefig("imag
```



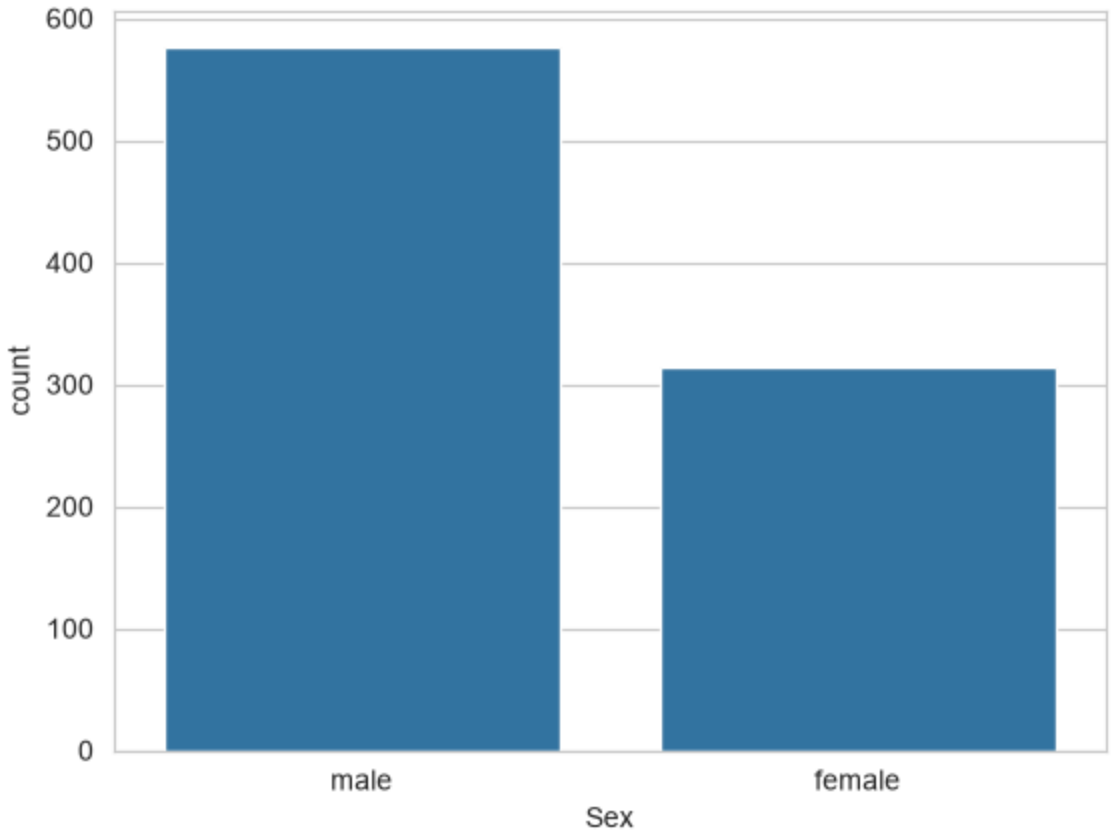
In [52]:
```
sns.countplot(x="Survived", data=df); plt.savefig("images/survived_count.png"); plt.s
```



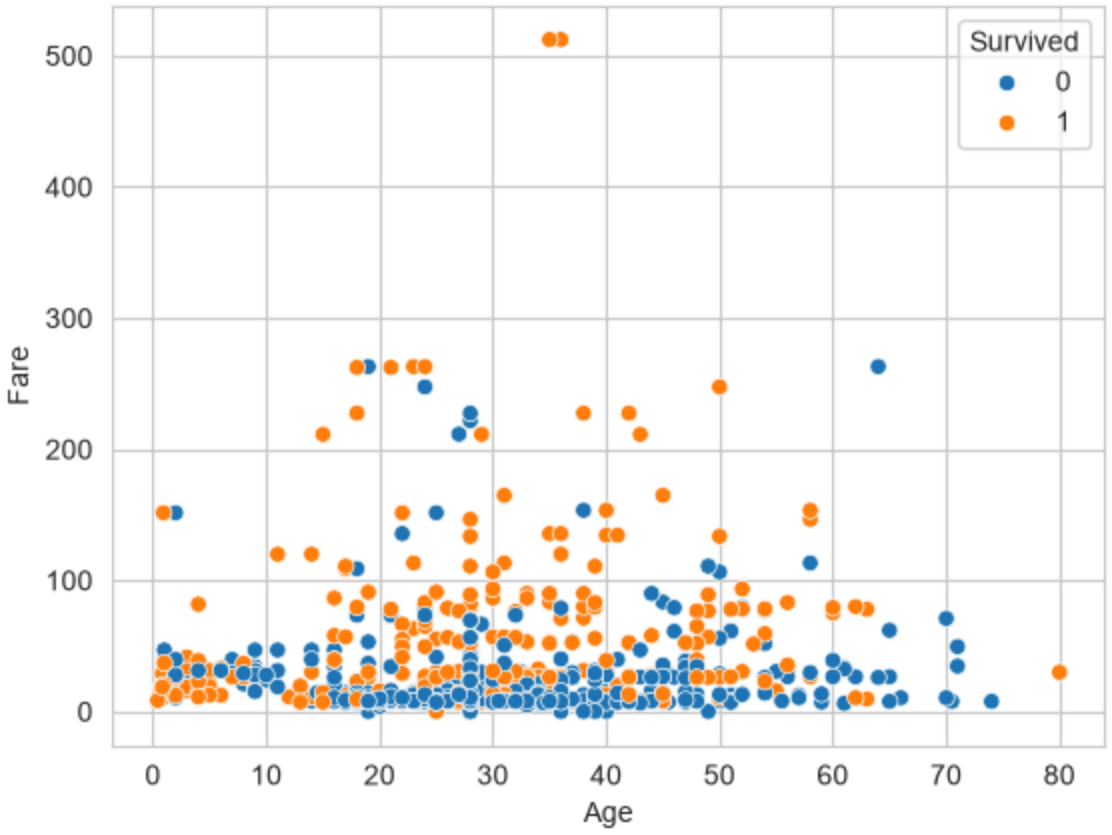
In [53]:
```
sns.countplot(x="Pclass", data=df); plt.savefig("images/pclass_count.png"); plt.show(
```



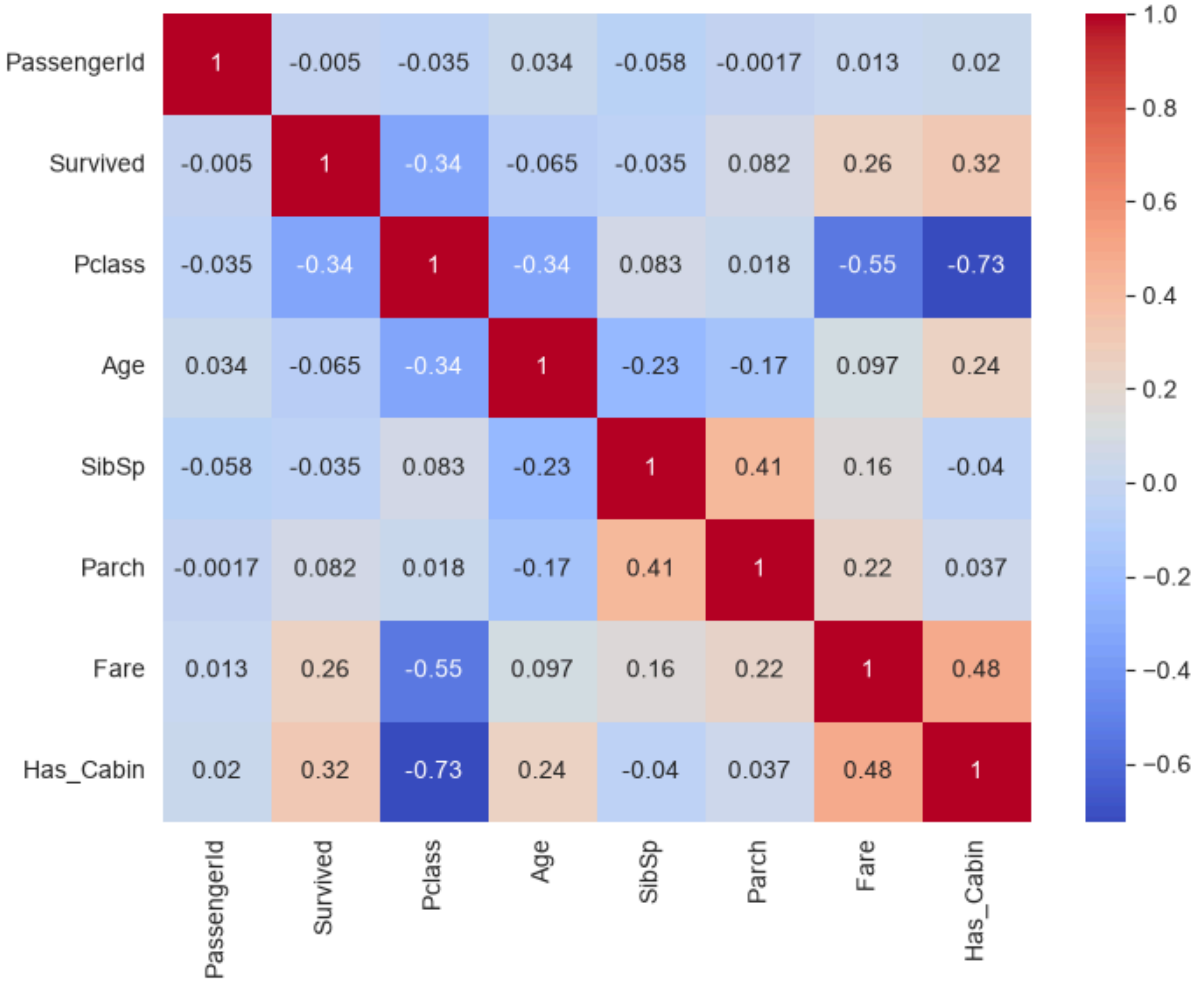
In [54]:
```
sns.countplot(x="Sex", data=df); plt.savefig("images/sex_count.png"); plt.show()
```



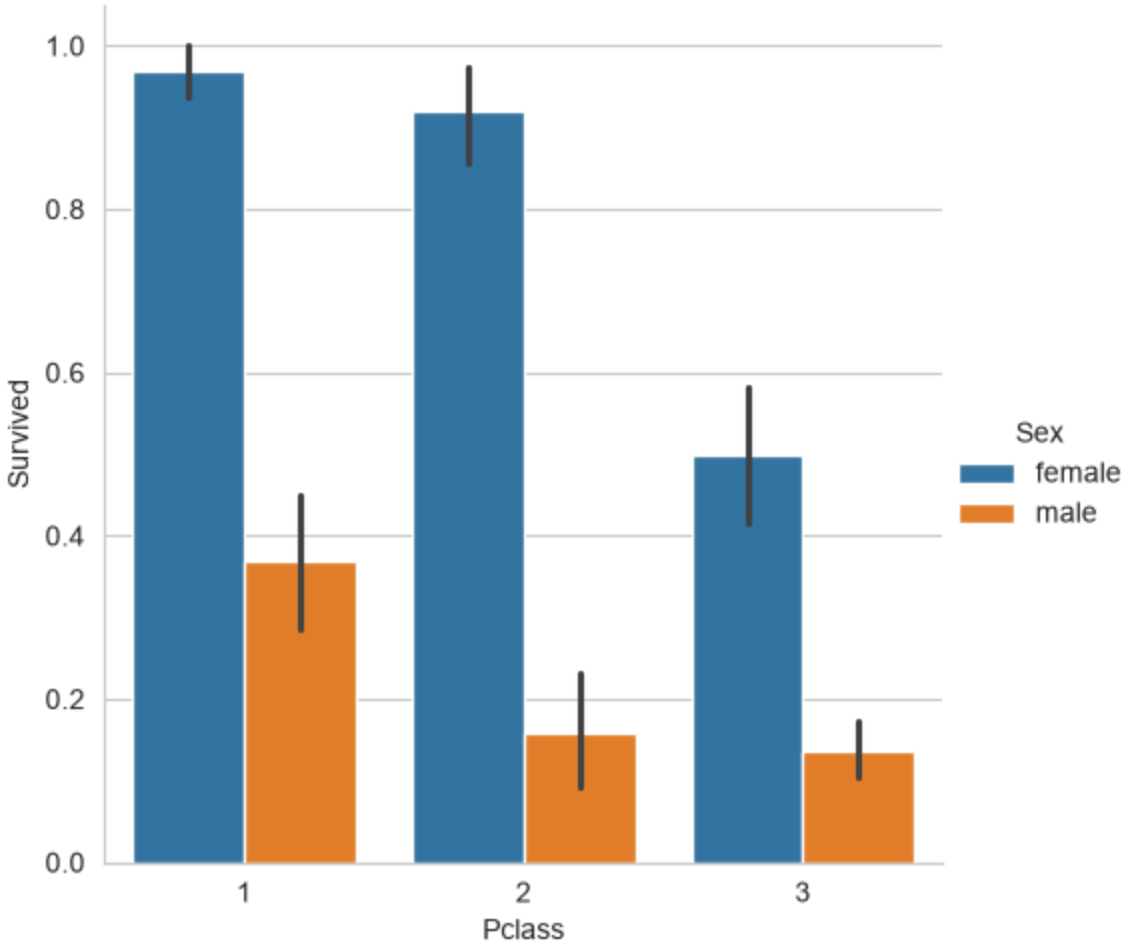
In [55]:
```
sns.scatterplot(x="Age", y="Fare", hue="Survived", data=df); plt.savefig("images/scat
```



In [56]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df.select_dtypes("number").corr(), annot=True, cmap="coolwarm")
plt.savefig("images/heatmap.png"); plt.show()
```



In [57]:
```
sns.catplot(kind="bar", x="Pclass", y="Survived", hue="Sex", data=df)
plt.savefig("images/surv_class_sex.png"); plt.show()
```



In [ ]: